SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: MLA0202 COURSE NAME: Fundamentals of Machine Learning

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COURSE OUTCOME COVERED

CO2: *Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications.* *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 10 **Total Marks:50**

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I **Kruthika Sre S / 192525360** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
Kruthika Sre

Annexure A

Debugging & Optimisation Task

Q1. A real estate company developed a linear regression model to predict house prices, but the model is showing high prediction error.

(i) Identify possible causes for the high error (e.g., multicollinearity, outliers, poor feature selection).

(ii) Analyze the dataset/model behavior to detect these issues.

(iii) Apply suitable optimization techniques (such as feature scaling, removing outliers, or adding relevant features) to improve the model performance. (5 Marks-8 Mins)

Q2.An email classification system using a linear model is incorrectly classifying several spam emails as non-spam.

(i) Identify the reasons for misclassification (e.g., improper decision boundary, imbalanced data).

(ii) Examine the current decision boundary and classification errors.

(iii) Refine the model using optimization techniques such as adjusting thresholds, feature engineering, or re-training with balanced data.(5 Marks-8 Mins)

Q3.A Naïve Bayes classifier is giving poor accuracy due to missing or zero probability values in the dataset.

(i) Identify the issue related to missing or zero probabilities.

(ii) Analyze how these values affect classification results.

(iii) Apply appropriate debugging and optimization methods such as Laplace smoothing to improve accuracy. (5 Marks-8 Mins)

Q4.A discriminant function is unable to clearly separate two classes, leading to incorrect classifications.

(i) Identify possible reasons for poor class separation (e.g., overlapping features, incorrect weights).

(ii) Analyze the decision boundary and feature contribution.

(iii) Improve the discriminant function using optimization strategies such as adjusting weights, feature transformation, or normalization.(5 Marks-8 Mins)

Q5.A Bayesian logistic regression model developed for customer churn prediction is suffering from overfitting, resulting in poor generalization.

(i) Identify signs and causes of overfitting in the model.

(ii) Analyze model performance on training vs testing data.

(iii) Apply debugging and regularization techniques (such as L1/L2 regularization or cross-validation) to improve model performance.(5 Marks-8 Mins)

Q6.A linear regression model built for predicting sales is performing poorly because the features (e.g., advertising budget in thousands and number of outlets in units) are on different scales.

(i) Identify how feature scaling issues affect model performance.

(ii) Analyze the impact of unscaled features on regression coefficients.

(iii) Apply appropriate optimization techniques such as normalization or standardization to improve the model.(5 Marks-8 Mins)

Q7.A binary classification model for fraud detection is biased towards the majority class, leading to poor detection of fraudulent cases.

(i) Identify the problem caused by class imbalance.

(ii) Evaluate model performance using suitable metrics (precision, recall, F1-score).

(iii) Apply optimization techniques such as resampling (oversampling/undersampling) or class weighting to improve performance.(5 Marks-8 Mins)

Q8.A Naïve Bayes classifier is underperforming because some input features are highly correlated, violating the independence assumption.

(i) Identify how correlated features affect Naïve Bayes performance.

(ii) Analyze feature relationships in the dataset.

(iii) Apply optimization techniques such as feature selection or dimensionality reduction to improve accuracy. (5 Marks-8 Mins)

Q9.A logistic regression model is unable to clearly separate two classes due to overlapping feature distributions.

(i) Identify the cause of overlapping decision boundaries.

(ii) Analyze the feature space and classification errors.

(iii) Improve the model using techniques such as feature transformation, polynomial features, or regularization. (5 Marks-8 Mins)

Q10.A discriminant analysis model is too simple and fails to capture the complexity of the dataset, resulting in low accuracy.

(i) Identify the signs of underfitting in the model.

(ii) Analyze model bias and feature limitations.

(iii) Apply optimization strategies such as adding features, increasing model complexity, or using nonlinear transformations. (5 Marks-8 Mins)

1. Improving a Linear Regression Model for House Price Prediction

(i) Possible Causes of High Prediction Error

High prediction error may occur due to multicollinearity, outliers, irrelevant features, insufficient data, or a nonlinear relationship between features and house prices. Multicollinearity occurs when independent variables are highly correlated.

(ii) Analyzing the Dataset and Model Behavior

The dataset can be examined for missing values, outliers, and correlations between features. A correlation matrix helps identify multicollinearity. Residual plots and Mean Squared Error (MSE) help identify prediction errors. A large difference between actual and predicted prices indicates poor model performance.

(iii) Optimization Techniques

Feature scaling can bring numerical features to comparable scales. Outliers can be treated using the IQR method after verifying their validity. Irrelevant features should be removed, and useful features such as location, house size, and number of bedrooms can be added. These improvements can reduce prediction error and increase model accuracy.

2. Improving an Email Spam Classification System

(i) Reasons for Misclassification

Spam emails may be classified as non-spam due to an unsuitable decision threshold, irrelevant features, insufficient training data, or class imbalance. Poor feature selection may also prevent the model from identifying important spam characteristics.

(ii) Examining the Decision Boundary and Errors

The model's predicted labels can be compared with actual labels using a confusion matrix. Spam emails incorrectly classified as non-spam are false negatives. The decision boundary should be examined to determine whether the model is too strict when identifying spam. Precision, recall, and F1-score can help evaluate performance.

(iii) Refining the Model

The classification threshold can be lowered to identify more spam emails. Features such as suspicious links, keywords, sender information, and email frequency can be included. The model can be retrained using balanced data or class weights. These techniques can improve spam detection and reduce false negatives.

3. Improving a Naïve Bayes Classifier Using Laplace Smoothing

(i) Identifying the Issue

A Naïve Bayes classifier may perform poorly when some feature values have zero probabilities or when the dataset contains missing values. A zero probability for a feature can make the entire class probability zero because Naïve Bayes multiplies conditional probabilities.

(ii) Analyzing the Effect on Classification

The classifier calculates class probabilities using prior and conditional probabilities. If one conditional probability becomes zero, the product may become zero regardless of the other feature values. Missing data can also affect probability estimates if it is not handled appropriately, resulting in incorrect classifications.

(iii) Debugging and Optimization Methods

Laplace smoothing can solve the zero-probability problem by adding a small count to each possible feature value. The smoothed probability is:

$P(x|C) = (Count(x,C) + 1) / (Count(C) + k)$

Here, Count(x,C) is the number of occurrences of feature value x in class C, Count(C) is the total number of relevant observations in class C, and k is the number of possible feature values.

Missing values should be handled using suitable imputation methods or by appropriately excluding unavailable feature information. These methods improve probability estimation and classification performance.

4. Improving a Discriminant Function for Class Separation

(i) Possible Reasons for Poor Class Separation

Poor class separation may occur due to overlapping features, incorrect weights, irrelevant variables, or features with different scales. A linear discriminant function may also perform poorly when the relationship between the classes is nonlinear.

(ii) Analyzing the Decision Boundary and Feature Contribution

The decision boundary can be examined by plotting the data points and identifying misclassified observations. The discriminant function is generally expressed as:

$g(x) = w_1x_1 + w_2x_2 + ... + w_nx_n + b$

Here, w represents feature weights, x represents input features, and b is the bias. The weights indicate the contribution of each feature to the decision. Incorrect weights or overlapping feature distributions can lead to classification errors.

(iii) Optimization Strategies

The weights and bias can be adjusted using suitable training or optimization methods. Feature normalization can reduce scale differences, while feature selection can remove irrelevant variables. Nonlinear transformations or additional informative features can improve class separation when a linear boundary is insufficient.

5. Reducing Overfitting in Bayesian Logistic Regression

(i) Signs and Causes of Overfitting

Overfitting occurs when a model learns the training data too closely, including noise and irrelevant patterns. It may result from excessive model complexity, too many features, or insufficient training data. A common sign is high training accuracy but low testing accuracy.

(ii) Analyzing Training and Testing Performance

The model's performance should be evaluated separately on training and testing datasets. If training error is low but testing error is high, the model is likely overfitting. Cross-validation can provide a more reliable estimate of how well the model generalizes to unseen data.

(iii) Debugging and Regularization Techniques

L1 regularization can reduce unnecessary feature coefficients, while L2 regularization penalizes large coefficients. Cross-validation can help select suitable regularization parameters. Reducing irrelevant features and collecting more training data can also improve generalization. In Bayesian logistic regression, suitable prior distributions can help control coefficient values and reduce overfitting.

6. Improving Linear Regression Through Feature Scaling

(i) Effects of Feature Scaling Issues

Feature scaling is important when input features have significantly different numerical ranges. For example, advertising budget may be measured in thousands, while the number of outlets is measured in individual units. Such differences can affect the numerical behavior and convergence speed of gradient-based optimization algorithms.

(ii) Impact on Regression Coefficients

Features with different scales may produce coefficients with different magnitudes. This does not necessarily make ordinary least squares regression inaccurate, but it can make coefficient magnitudes difficult to compare. Unscaled features may also cause slow or unstable convergence when gradient descent is used.

(iii) Optimization Techniques

Normalization scales values to a fixed range, usually between 0 and 1. Standardization transforms features to have a mean of 0 and a standard deviation of 1.

Standardization formula:

$z = (x - \mu) / \sigma$

Here, x is the original value, μ is the mean, and σ is the standard deviation.

Applying a suitable scaling method can improve optimization speed and numerical stability. The same scaling parameters must be applied to new data before making predictions.

7. Improving Fraud Detection with Imbalanced Data

(i) Identifying the Class Imbalance Problem

Class imbalance occurs when one class contains significantly more observations than another. In fraud detection, legitimate transactions usually outnumber fraudulent transactions. Consequently, a model may predict most transactions as legitimate and fail to detect fraud.

(ii) Evaluating Model Performance

Accuracy alone is insufficient for evaluating an imbalanced classification model. Precision, recall, and F1-score provide more useful information.

Precision = TP / (TP + FP)

Recall = TP / (TP + FN)

F1-score = 2 × (Precision × Recall) / (Precision + Recall)

Here, TP represents true positives, FP represents false positives, and FN represents false negatives. Recall is especially important when detecting fraudulent transactions because it measures the proportion of actual fraud cases identified correctly.

(iii) Optimization Techniques

Oversampling increases the representation of the minority class, while undersampling reduces the number of majority-class observations. Class weighting assigns greater importance to fraudulent cases during training. The classification threshold can also be adjusted to improve fraud detection. The final model should be evaluated using precision, recall, F1-score, and a confusion matrix.

8. Improving Naïve Bayes with Correlated Features

(i) Effects of Correlated Features

Naïve Bayes assumes that features are conditionally independent given the class. Highly correlated features violate this assumption. When similar features provide overlapping information, the classifier may effectively count the same evidence multiple times, leading to inaccurate probability estimates.

(ii) Analyzing Feature Relationships

A correlation matrix can identify highly correlated numerical features. Scatter plots and pairwise comparisons can also reveal relationships between variables. The selected features should be examined to determine whether they provide independent and useful information for classification.

(iii) Optimization Techniques

Feature selection can remove redundant or highly correlated variables. Dimensionality reduction methods, such as Principal Component Analysis (PCA), can transform the original features into a smaller set of components. However, PCA does not guarantee conditional independence. The classifier should be retrained and evaluated using suitable metrics to determine whether these changes improve accuracy.

9. Improving Logistic Regression for Overlapping Classes

(i) Identify the Cause of Overlapping Decision Boundaries

Logistic regression may struggle when the feature distributions of two classes overlap significantly. Some observations from different classes may have similar feature values, making them difficult to distinguish using a linear decision boundary. Irrelevant features and noisy data can further increase classification errors.

(ii) Analyze the Feature Space and Classification Errors

A scatter plot can show the distribution of the two classes and reveal their degree of overlap. A confusion matrix identifies false positives and false negatives. Precision, recall, and F1-score can help determine which types of classification errors are most frequent.

(iii) Improve the Model Using Optimization Techniques

Polynomial features can help logistic regression represent nonlinear relationships. Feature transformation can make class patterns easier to distinguish, while feature selection can remove irrelevant variables. Regularization can prevent overfitting when additional features are introduced. The improved model should be evaluated using cross-validation and appropriate classification metrics.

10. Improving an Underfitting Discriminant Analysis Model

(i) Identify the Signs of Underfitting

Underfitting occurs when a model is too simple to capture the underlying patterns in the dataset. It generally produces high training and testing errors. In discriminant analysis, this may happen when the model's assumptions are unsuitable for the data or important features are missing.

(ii) Analyze Model Bias and Feature Limitations

Underfitting is associated with high bias. The model may fail to capture nonlinear relationships or differences between the classes. Training and testing performance should be compared, and feature distributions should be examined to identify missing information or

unsuitable model assumptions.

(iii) Apply Optimization Strategies

Relevant features can be added to improve the model's ability to distinguish between classes. Nonlinear transformations can help capture more complex patterns. A more flexible classification model may be selected if the existing discriminant analysis method is too restrictive. The revised model should be evaluated using cross-validation to ensure that the increased complexity improves generalization.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation